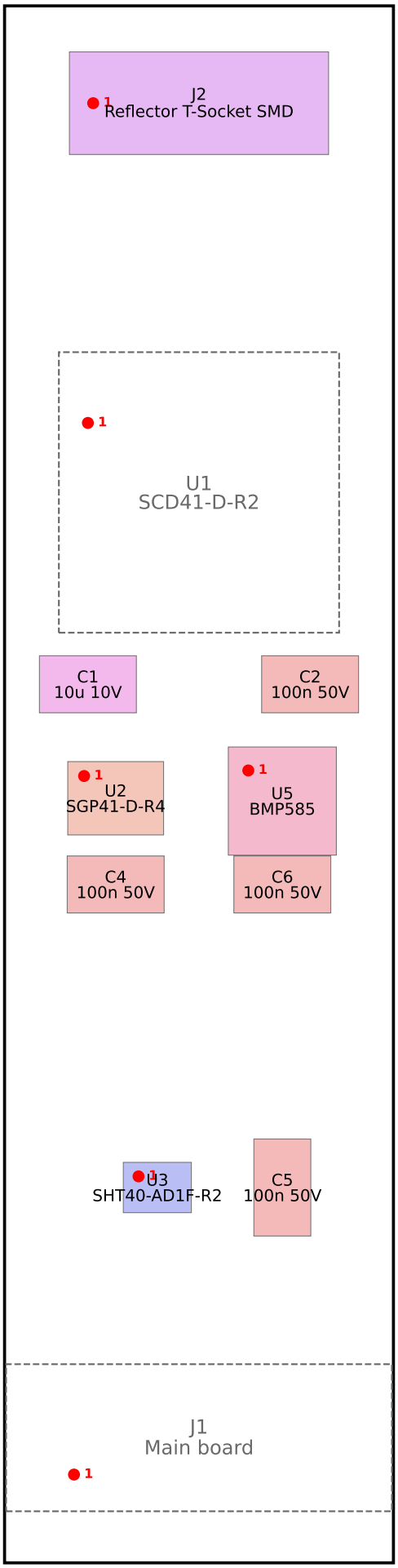
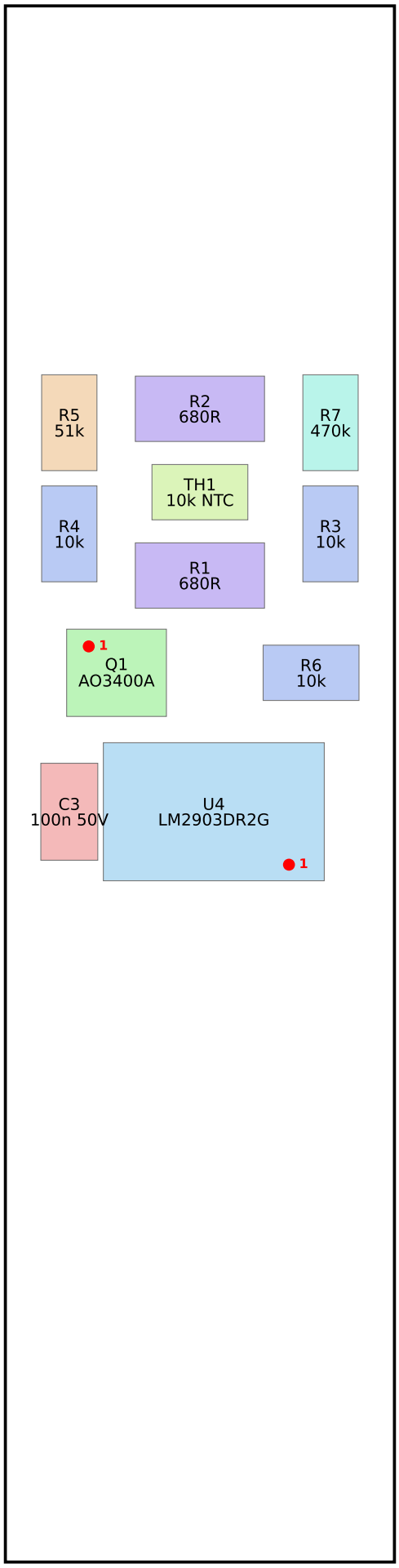
probe - TOP

red dot 1 = pin 1 black bar = cathode red + = cap + orange corner = hand-fitted dashed = fit last (key: last page)



probe - BOTTOM (seen from below, mirrored)

red dot 1 = pin 1 black bar = cathode red + = cap + orange corner = hand-fitted dashed = fit last (key: last page)



probe - values

- 100n 50V x4 top: C2, C4, C5, C6
- 100n 50V x1 bottom: C3
- 10k x3 bottom: R3, R4, R6
- 10k NTC x1 bottom: TH1
- 10u 10V x1 top: C1
- 470k x1 bottom: R7
- 51k x1 bottom: R5
- 680R x2 bottom: R1, R2
- AO3400A x1 bottom: Q1
- BMP585 x1 top: U5
- LM2903DR2G x1 bottom: U4
- Main board x1 top: J1
- Reflector T-Socket SMD x1 top: J2
- SCD41-D-R2 x1 top: U1
- SGP41-D-R4 x1 top: U2
- SHT40-AD1F-R2 x1 top: U3

probe - key

Marks on the board pages

red dot + 1 : pin 1 of ICs, transistors, LEDs, optocouplers, connectors and modules

black bar   : diode cathode, matching the band printed on the diode body

red +   : + pad of electrolytic / tantalum capacitors

no mark   : resistors, ceramic capacitors, inductors, fuses, crystals, pin strips, bidirectional TVS

orange corner: part fitted by hand (not placed by JLCPCB); dashed outline: large part, fit it last

the BOTTOM page is mirrored: it shows the board as you see it from below

Finding pin 1 on the real part

SOT-23 (3 legs): the single leg on its own side is pin 3; pin 1 is left of it on the two-leg side

SOT-23-5 / SC-70 / SOIC / VSSOP / QFN / LGA: a dot, dimple or bevelled edge marks pin 1

Addressable LEDs: a marked (cut / green) corner - check the datasheet, it is easy to misread

Diodes: the band is the cathode (K)

Electrolytic capacitors: the stripe marks the - side (the other pad is +)

Tools

0805 / SOT-23: a 3-5x magnifier or a 10x loupe

QFN / LGA / DFN / SC-70: a USB microscope (check the pads under the part after reflow)